RDS (Relational Database Service)

What is DATA

1. In simple words, data can be facts related to any object.
2. For e.g: your age, job, house no, contact no., name, places are some data related to you.

What is DATABASE

1. Database is a systematic collection of data in a organise form.
2. Databases support storage and manipulation of data.
3. e.g.: Facebook, telecom companies, amazon.com

What is DBMS

DBMS is a collection of programs which enable its users to access database, manipulate data, reporting/ representation of data.

Types of DBMS

1. Hierarchical
2. Network
3. Relational
4. Object oriented

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Fields

1. **row/record/tuple**
   a row contains the specific information about the individual entry in the table
2. **column/attribute/field**
   a column stores a specific type of data (roll_no, name, branch)
3. **primary key**
   each table has a unique identifier known as primary key (example: - roll_no, account_no)
4. **schema**
   it defines the structure of the database

RELATIONAL DATABASE

1. A relational database is a data structure that allows you to link information from different tables of different types of data bucket.
2. Tables are related to each other.
3. All fields must be filled.
4. Best suited for OLTP (online transaction processing)
5. Relational DB: MySQL, Oracle, DBMS, IBM DB2
6. A row of a table is also called records. It contains the specific information of each individual entry in the table.
7. Each table has its own primary key.
8. A schema (design of database) is used to strictly define tables, columns, indexes and relation between tables.
9. Relational DB are usually used in enterprises application/scenario. Exception in MySQL which is used for web application.
10. Common application for MySQL includes php and java-based web applications that requires a database storage backend. E.g: JOOMLA
11. Cannot scale out horizontally.
12. Virtually all relational DB uses SQL.

static content ------> predefined Shema -------> vertical scalable

RDS is a fully managed AWS database services where AWS is responsible for…

1. security and patching
2. automated backups
3. software updates for the database engine
4. multi-AZ deployment (disaster recovery)

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NON-RELATIONAL DATABASE

1. Non-relational databases store data without a structured mechanism to link data from different tables to one another.
2. Required low-cost hardware.
3. Much faster performance (read/write) compared to relational DBs.
4. Horizontal scaling is possible.
5. Never provide tables with flat fixed column records. It means schema-free.
6. Best suited for online analytical processing (OLAP).
7. E.g: of NoSQL databases: MongoDB, Casssandra, DynamoDB, Postegre, Raven,Redis.

unstructured data ----------> dynamic schema --------> horizontality scalable

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Types of No-SQL Databases

1. Columnar Database (cassandra, HBase)
2. Document Databse (MongoDB, CouchDB, RavenDB)
3. Key Value Database (Redis, Riak, DynamoDB, Tokyo Cabinet)
4. Graph Based Database (Neo4J, FlockDB)

1. Columnar Database:

1. A columnar database is a DBMS that stores data in columns instead of Rows.
2. In a columnar DB all the column-1 values are physically together followed by all the column-2 values.
3. In a row-oriented DBMS, the data would be stored like this: (1, bob, 30, 8000: 2, arun, 26, 4000: 3, vian, 39, 2000 ;)
4. In a column-based DBMS the database would be stored like this: (1, 2, 3: bob, arun, vian; 30, 26, 39; 8000, 4000, 2000 ;)
5. Benefit is that because a column-based DBMS is self-indexing, it uses less disk space that a RDBMS containing the same data. It easily performs operation like min, max, average.

2. Document Database:

1. Document DB make it easier for developer to store and querying data in a DB by using the same document model format they use in their application code.
2. Document DBs are efficient for storing catalogue.
3. Store semi-structure data as document typically in JSON or XML format. (example)
4. A document database is a great choice for contain management application such as blogs and video platform.

3. Key-Value Database:

1. A key-value DB is a simple DB that uses an associative array (like dictionary) as a fundamental model where each key is associated with one and only one value in a collection.
2. It allows horizontal scaling.
3. Used cases: shopping cart, and session store in app like fb and twitter.
4. They improve application performance by storing critical pieces of data in memory for low latency access.
5. Amazon elastic cache as an in-memory key-value stores.

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4. Graph Based Database:

1. A graph DB is basically a collection of nodes and edges.
2. Each node represents an entity and each edges represent a connection or relationship between two nodes.
3. In an AWS fully managed relational DB engines service where AWS is responsible for:
    1. Security and patching.
    2. Automated backup.
    3. Software updates for DB engine.
    4. If selected multi-AZ with synchronous replication between the active and stand by DB instances.
    5. Automatic failover if multi-AZ option was selected.
    6. By default, every DB has weekly maintenance window. (Max 35 days.) Settings managed by the users:
    7. Managing DB settings.
    8. Creating relational database schema.
    9. Database performance tuning.

Relational Database Engine Options

1. aurora (MySQL)
2. Oracle
3. MySQL
4. MariaDB
5. Postgre SQL
6. Microsoft SQL
7. IBM db2
8. aurora (PostgreSQL)

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There are two Licensing Options

1. BYOL (Bring Your Own License)
2. License from AWS on hourly basis

RDS Limits

1. 10 of this 40 can be Oracle or MS-SQL server under license included model. Or
2. Under BYOL model, all 40 can be any DB engine you need. RDS Instance Storage:
3. Amazon RDS use EBS volumes (not instance store) for DB and logs storage.

1. General Purpose: use for DB workloads with moderate I/O requirement.
Limits: min: 20 GB
Max: 16384 GB

2. Provisional IOPS RDS Storage: use for high performance OLTP workloads.
Limits: min: 100GB
Max: 16384GB

Templates available in RDS

a. Production
b. Dev/Test
c. Free-Tier

DB Instance Size

a. Standard class
b. Memory-Optimized class
c. Burstable class

What is Multi-AZ in RDS

1. You can select multi-AZ option during RDS DB instance launch.
2. RDS service creates a standby instance in a different AZ in the same region and configure “synchronous replication” between the primary and standby.
3. You cannot read/write to the standby RDS DB instances.
4. You cannot select which AZ in the region will be chosen to create the standby DB instance.
5. You can however view, which AZ is selected after the standby is created.
6. Depending on the instance class it may take 1 to few minutes to failover to the standby instance.
7. AWS recommends the use of provisioned IOPS instances for multi-AZ RDS instance.

When Multi-AZ RDS Failover Triggers

1. In case of failure of primary DB instance failure.
2. In case of AZ failure.
3. Loss of network connectivity to primary DB.
4. Loss of primary EC2 instance failure.

5. EBS failure of primary DB instance.
6. The primary DB instance is changed.
7. Patching the O.S of the DB instance.
8. Manual failover. (in case of rebooting.)

Multi-AZ RDS Failover Consequences

1. During failover the CNAME of the RDS DB instance is updated to map to the standby IP address.
2. It is recommended to use the end point to reference your DB instances and not its IP address.
3. The CNAME doesn't change because the RDS endpoint doesn't change.
4. RDS end point doesn't change by selecting multi-AZ option, however the primary and standby instances will have different IP addresses, as they are in different AZ.
5. It is always recommended that you do not use the IP address to point RDS instances, always use endpoint. By using endpoint there will be no change whenever a failover happens.

When we do manual failover

1. In case of rebooting.
2. This is by selecting the "reboot with failover" reboot options on the primary RDS DB instances.
3. A DB instance reboot is required for changes to take effect when you change the DB parameter group on when you change a static DB parameter.
4. Whenever failover occurs AWS RDS sends SNS notification.
5. You can use API calls to find out the RDS events occurred in the last 14 days.
6. Even you can use CLI to view last 14 days events.
7. Using AWS console you can only last one day events
8. In case of OS patching, system upgrades and DB scaling, these things happen on standby first then on primary to avoid outage.
9. In multi-AZ, snapshots and automated backups are done on standby instance to avoid I/O suspension on primary.

RDS Multi-AZ Deployment Maintenance

1. Firstly, perform maintenance of standby.
2. Now convert standby into primary so that maintenance can be done on primary. (currently)

3. You can manually upgrade a DB instance to a supported DB engine version from AWS console as follows: - RDS->DB instance->modify DB->set DB engine version.
4. By default, change will take effect during the next maintenance window.
5. Or you can force a immediate upgrade if you want.
6. In multi-AZ version upgrade will be conducted by both primary and standby at the same time which will cause an outage.
7. Do it during maintenance window.

There are two methods to backup and restore your RDS DB instances

1. AWS RDS automated backup
2. User initiated manual backup

1. Either you can take backup of entire DB instance or just the DB.
2. You can create a restore volume snapshots of your entire DB instances.
3. Automated backups by AWS, backup your DB data to multiple AZ to provide for data durability.
4. Select-automated backup in AWS console.
5. Stored in Amazon S3.
6. Multi-AZ automated backups will be taken from the standby instance.
7. The DB instance must be in “ACTIVE” state for automated backup.
8. RDS automatically backups the DB instances daily by creating a storage volume snapshot of your DB instance (fully daily snapshots) including the DB transaction logs.
9. You can decide when you would like to take backup (window)
10. No additional charge for RDS backing up your DB instance.
11. For multi-AZ deployment, backups are taken from the standby DB instance (true for Maria DB, MySQL, Oracle, Postgre SQL).
12. Automated backups are deleted when you delete your RDS DB instance.
13. An outage occurs if you change the backup retention period from zero to non-zero value or the other way around.
14. Retention period of automate backup is 7 days (by default) via AWS console.
15. AWS Aurora is an exception. Its default is 1 day.
16. Via CLI or API 1 day by default.
17. You can increase it up to 35 days.
18. If you don’t want backup, put zero “0” in the retention period.
19. In case of manual snapshot, point-in-time recovery is not possible.
20. Manual snapshot is also stored in S3.

21. They are not deleted automatically, if you delete RDS instance.
22. Take a final snapshot before deleting your RDS DB instance.
23. You can share manual snapshot directly with other AWS Account.
24. When you restore a DB instance only the default DB parameters and security groups are associated with the restored instance.
25. You cannot restore a DB snapshot into an existing DB instance rather it has to create a new DB instance it has new endpoint.
26. Restoring from the backup or a DB snapshot changes the RDS instance endpoint.
27. At the time of restoring, you can change the storage type (general purpose or provisioned.)
28. You cannot encrypt an existing unencrypted DB instance.
29. To do that you need to: create a new encrypted instance and migrate your data to it (from unencrypted to encrypted) or you can restore from a backup/snapshot into a new encrypted RDS instance.
30. RDS supports encryption-at-rest for all DB engines using KMS.
31. What actually encrypted when data-at-rest:
    a) All its snapshots.
    b) Backups of DB (S3 storage.)
    c) Data on EBS volume.
    d) Read replica created from the snapshots.

Some points related to RDS Billings:

1. No upfront cost.
2. You have to pay only for:
    a) DB instance hours (partial hour charged as full hours)
    b) Storage GB/month.
    c) Internet data transfer.
    d) Backup storage (i.e S3)

This is increased by increasing DB backup's retention period.

Also charged for

a) Multi-AZ DB hours.
b) Provisioned stage (multi-AZ)
c) Double write I/O

d) You are nor charged for DB data transfer during replication from primary to standby.

Database backup can be taken in two main ways:

1.Automated Backup

1. This is automatic (system does it for you)
2. Scheduled daily (or based on settings)
3. Keeps backups for a retention period (e.g., 7 days)
4. Includes:
   1.Full backup
   2.Transaction logs (for point-in-time recovery)
5. Example:
   In Amazon Web Services RDS → you enable automated backup → it runs daily without your action

2.Manual Snapshot

1. This is taken by you manually
2. 2.You decide when to take it
3. 3.Stored until you delete it
4. 4.Used before risky changes (updates, migrations)
5. Example:
   Before upgrading database → you take a snapshot

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